

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	4 Marks

SMART METRO TICKET GENERATING SYSTEM

Data Flow Diagrams and User Stories

1. Purpose

This document presents the Data Flow Diagrams (DFDs) that model how information moves through the Smart Metro Ticket Generating System, together with the user stories that translate system requirements into concrete, testable statements of commuter and staff needs. DFDs are presented at two levels of detail: Level 0 (Context Diagram), which shows the system as a single process bounded by its external actors, and Level 1, which decomposes that process into its major internal sub-processes and data stores.

1.1 DFD Level 0 — Context Diagram

The context diagram establishes the system boundary. Four external entities interact with the Smart Metro Ticket Generating System: the Commuter (initiates ticket requests and pays), the Payment Gateway (authorizes transactions), Station Staff (handle exceptions at gates), and the Metro Administrator (configures fare rules and consumes reports).

Smart Metro Ticket Generating System — DFD Level 0 (Context Diagram)

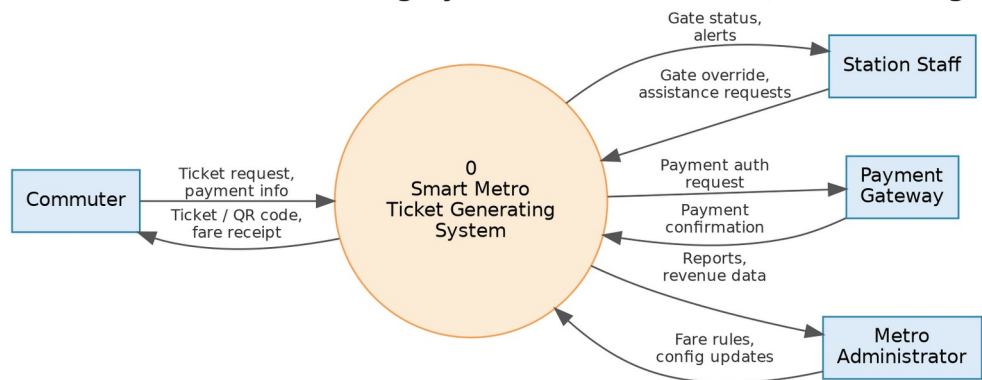


Figure 1: DFD Level 0 — Context diagram for the Smart Metro Ticket Generating System

2. DFD Level 1 — Detailed Process Decomposition

Level 1 decomposes the single context-level process into five core sub-processes, each reading from or writing to one of four data stores. This view is the primary reference for engineering teams implementing individual microservices, as it makes explicit which store each process owns and where synchronous versus asynchronous hand-offs occur.

Smart Metro Ticket Generating System — DFD Level 1 (Detailed)

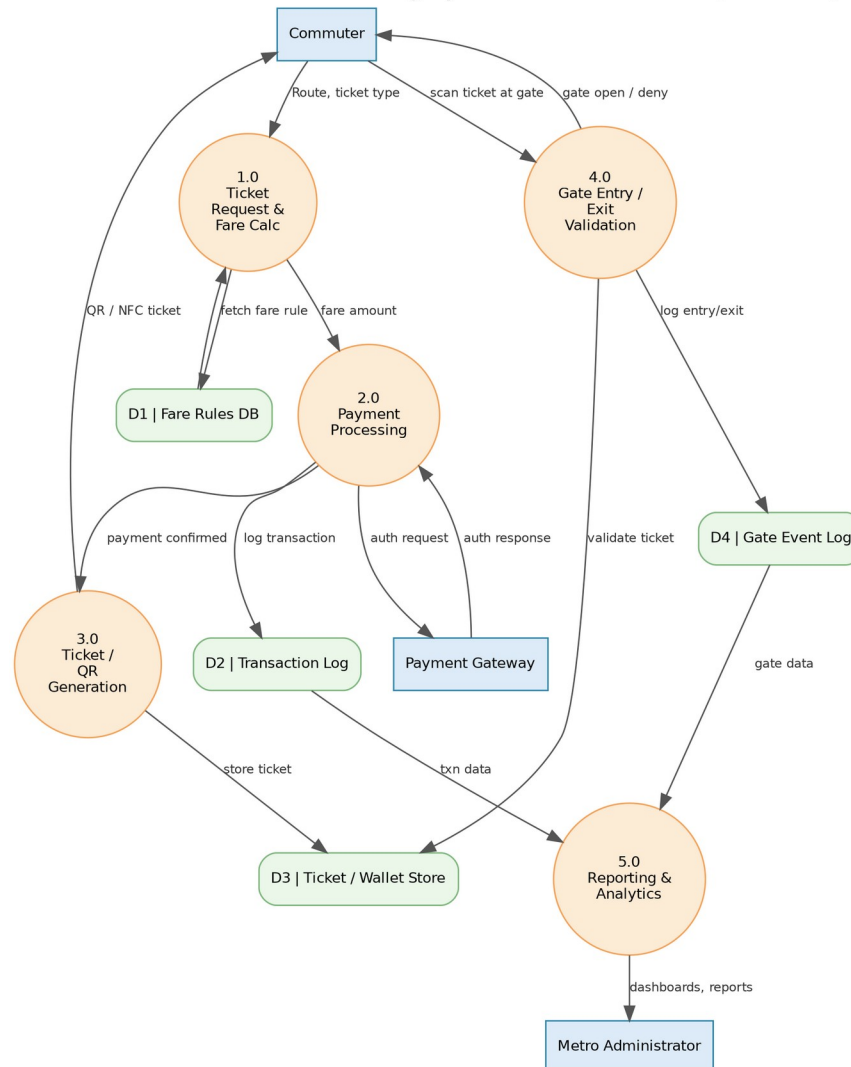


Figure 2: DFD Level 1 — Detailed data flow across ticket request, payment, issuance, gate validation and reporting processes

2.1 Process Descriptions

- 1.0 Ticket Request & Fare Calculation — receives route/ticket-type input from the commuter and computes fare using rules cached from D1 (Fare Rules DB).
- 2.0 Payment Processing — orchestrates authorization with the external Payment Gateway and writes an immutable record to D2 (Transaction Log).
- 3.0 Ticket / QR Generation — issues the QR code or updates the NFC smart card, persisting the ticket to D3 (Ticket / Wallet Store).
- 4.0 Gate Entry / Exit Validation — validates a scanned ticket against D3 in real time and logs every tap to D4 (Gate Event Log).
- 5.0 Reporting & Analytics — aggregates D2 and D4 data into dashboards and reports consumed by the Metro Administrator.

3. User Stories

The user stories below translate the functional requirements and data flows on the preceding pages into commuter- and staff-facing statements, each with a measurable acceptance criterion used during sprint planning and UAT sign-off.

ID	User Story	Acceptance Criteria
US-01	As a commuter, I want to buy a single-journey ticket in the app so I can travel without visiting a counter.	Given valid payment details, ticket/QR is issued within 3s and visible offline.
US-02	As a commuter, I want to see the fare before I pay so I can confirm the cost.	Fare is displayed based on selected origin/destination prior to payment confirmation.
US-03	As a commuter, I want to tap my phone at the gate so I don't need a physical ticket.	NFC/QR tap opens gate in <1s for a valid, unused ticket.
US-04	As a commuter, I want to recharge my smart card from home so I don't queue at the station.	Balance updates in the card/wallet within 60s of successful payment.
US-05	As station staff, I want to override a gate for a failed scan so commuters aren't blocked unfairly.	Override requires staff authentication and is logged with reason code.
US-06	As a metro administrator, I want to update fare rules centrally so all stations apply them instantly.	Rule change propagates to all gates within 5 minutes and is version-logged.
US-07	As a finance officer, I want daily revenue reports so I can reconcile takings across stations.	Report auto-generates daily, broken down by station and payment method.
US-08	As a commuter, I want a refund if my journey is cancelled before travel so I'm not charged unfairly.	Refund request before gate entry is processed within 24 hours to original payment method.

3.1 Traceability Note

Each user story maps to one or more DFD processes and, in turn, to a functional requirement ID defined in the Solution Requirements document (e.g., US-01 and US-03 trace to FR-01/FR-04 and FR-05 respectively, and are realized by DFD processes 1.0 through 4.0). Maintaining this chain from requirement to data flow to user story to test case ensures that no capability is implemented without a verifiable, testable origin.